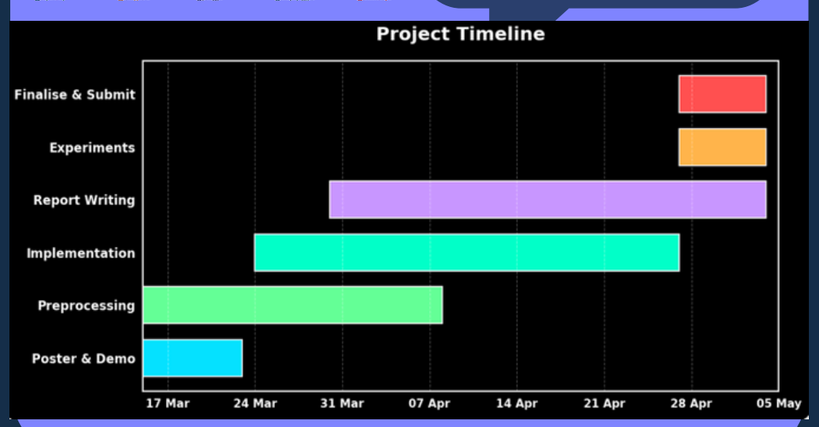
Scaling Trajectory Clustering on Apache Spark: Benchmarking TRACLUS, DBSCAN and K-Medoids on Real-World GPS Data

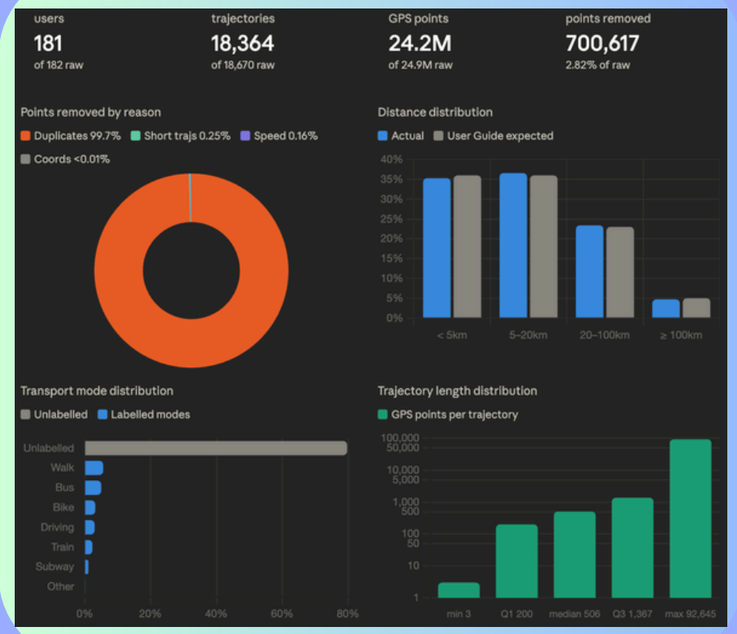
Research Questions

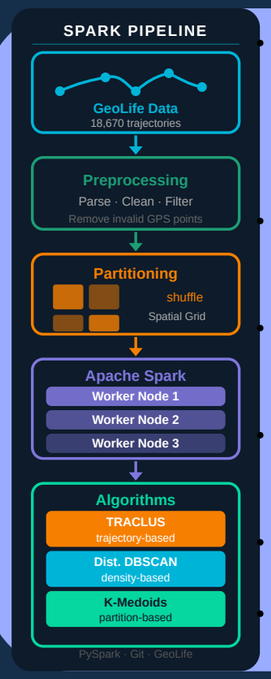
1. How does the runtime and resource usage of each algorithm (TRACLUS, Distributed DBSCAN, K-Medoids) scale as the dataset grows from the development subset to the full GeoLife dataset?
2. How does a local partition strategy (each worker cluster independently) compare to a global strategy (results merged across partitions) in terms of cluster quality and shuffle volume?
3. How well do the three algorithms recover meaningful mobility patterns (e.g., transport modes, common routes) from the GeoLife data, and how do quality metrics (silhouette score, noise ratio) vary across algorithms?

Timeline



Dataset: Microsoft GeoLife GPS Trajectories





References

- Ester et al. A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise. KDD 1996.
- Song et al. RP-DBSCAN: A Superfast Parallel DBSCAN Algorithm. SIGMOD 2018.
- Zaharia et al. Apache Spark: A Unified Engine for Big Data Processing. CACM 2016.
- Zheng et al. Mining Interesting Locations and Travel Sequences from GPS Trajectories. WWW 2009.
- Zheng et al. Understanding Mobility Based on GPS Data. UbiComp 2008.
- Zheng et al. GeoLife: A Collaborative Social Networking Service. IEEE DEB 2010.

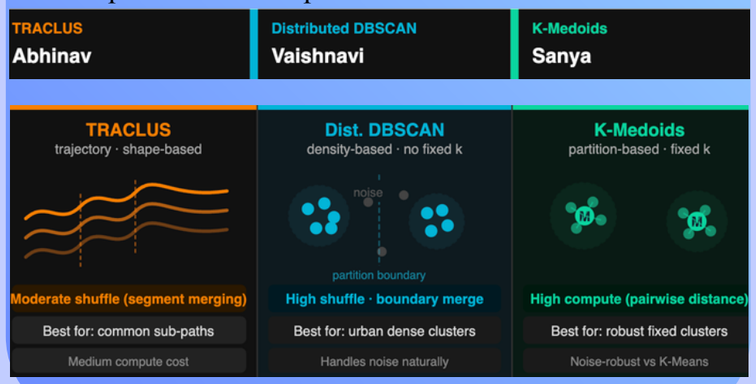
Clustering Algorithms

TRACLUS

- Partitions trajectories into line segments using the MDL principle, then clusters segments using a three-component distance measure (perpendicular, parallel, and angular).
- Identifies common sub-trajectory patterns rather than whole-trip clusters.
- The two-phase structure maps naturally onto Spark: partitioning is parallel, while the grouping phase requires cross-partition boundary merging.

Distributed DBSCAN

- Forms clusters by connecting points within a density threshold ε, with no predetermined **k** required.
- The dataset is spatially partitioned across workers, each running DBSCAN locally.
- The key distributed challenge is merging border clusters that span partition boundaries.
- Well-suited to GeoLife as urban mobility produces irregular dense clusters (e.g. around transport hubs) surrounded by sparse noise.

K-Medoids

- Clusters trajectories using actual data points as centres (medoids) rather than computed means, essential for trajectory data where an "average trajectory" has no meaningful interpretation.
- The main distributed challenge is computing the pairwise distance matrix (Hausdorff/DTW) across workers.
- More robust to noise than K-means and provides an interpretable fixed-k partition.